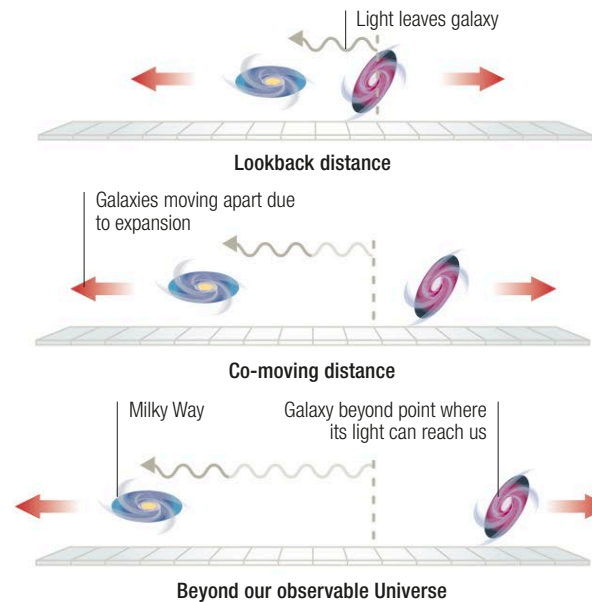


Lookback time

Although light is the fastest thing in the Universe, its speed is still limited, so that it crosses about 5.9 trillion miles (9.5 trillion kilometers) in a year. Coupled with cosmic expansion, this transforms our telescopes into time machines. The farther we look across space, the longer light has taken to reach us, and the farther back in time we are looking. Hence the most distant galaxies appear the most ancient and primitive.



◁ Stretching space

On cosmic scales, most of the redshift in distant galaxies arises not from pure Doppler shift but also from the way that light has stretched in its passage across expanding space.

◁ Receding galaxy

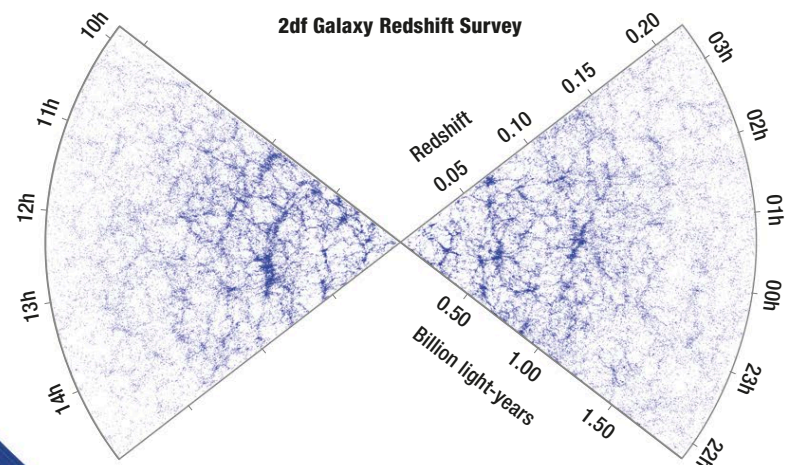
Although light may take a certain time to pass between distant galaxies, by the time it arrives the two galaxies may be much farther apart. The true separation of galaxies is called their co-moving distance.

◁ Shifted to invisibility

The most distant objects of all—the very first stars and galaxies formed in the early days of the Universe—have undergone extreme redshift, which renders them invisible to even the most advanced telescopes.

Mapping from redshift

Because more distant galaxies show larger redshifts (an effect called Hubble's Law), redshift itself can be used as a way of estimating distance for large numbers of galaxies that are too distant to measure in other ways. Mapping the redshifts of galaxies in different parts of the sky shows how galaxy superclusters form a web of elongated chains and sheets, known as filaments, around vast and apparently empty regions, called voids.



△ Redshift Survey

This map from the 2dF Galaxy Redshift Survey plots the redshifts of more than 200,000 galaxies across two broad swathes of sky, revealing structures that are hundreds of millions of light-years across. These structures are far too large to have formed by gravitational effects since the Big Bang. Instead they are thought to have been ingrained in the initial distribution of matter from which everything formed.